

Causal Survival - RHC Application

2026-05-31

Package Imports

EDA

```
library("Hmisc")  
library("tableone")  
library("xtable")  
library("ggplot2")  
library("survey")  
library("cobalt")
```

Estimation

```
library("survival")  
library("casebase")  
library("sandwich")  
library("foreach")  
library("doParallel")
```

Data Preparation

```
## Set random seed for reproducibility
s = 123

## Load the Right Heart Catheterization (RHC) dataset
getHdata(rhc)
```

Primary Analysis Variables

```
## Define follow-up time as the time from hospital admission to the last
## available contact or death date
time <- as.numeric(
  pmax(rhc$dthdte, rhc$lstctdte, na.rm = TRUE) - rhc$sadmdte
)

## Define the binary event indicator:
## 1 = death, 0 = censored
Delta <- ifelse(
  rhc$death == "Yes",
  1,
  0
)

## Define the binary treatment indicator:
## 1 = right heart catheterization (RHC) performed, 0 = not performed
A <- ifelse(
  rhc$swang1 == "RHC",
  1,
  0
)
```

Baseline Covariates

```
### Demographic and Socioeconomic Characteristics

## Patient sex
Gender <- rhc$sex

## Patient age at baseline
Age <- as.numeric(rhc$age)

## Patient weight in kilograms
## Note: The original dataset encodes some missing values as 0. These values
## are retained because replacing them with NA substantially alters the
## estimated propensity scores and covariate balance following weighting.
Weight <- as.numeric(rhc$wtkilo1)

## Patient race
Race <- droplevels(rhc$race)

## Educational attainment
Education <- as.numeric(rhc$edu)

## Income category
Income <- rhc$income

## Insurance classification
InsuranceClass <- rhc$ninsclas

### Pre-existing Comorbidities

## Names of baseline comorbidity variables to be recoded
hx.vars <- c(
  "chfhx",
  "cardiohx",
  "dementhx",
  "psychhx",
  "chrpulhx",
  "renalhx",
  "liverhx",
  "gibbledhx",
  "malighx",
  "immunhx",
  "transhx",
  "amihx"
)

## Convert labelled comorbidity variables to binary factors:
## "No" = absence of comorbidity, "Yes" = presence of comorbidity
rhc[hx.vars] <- lapply(rhc[hx.vars], function(x) {
  x <- as.numeric(x)

  factor(
```

```

    ifelse(x == 1, "Yes", "No"),
    levels = c("No", "Yes")
  )
})

## Extract recoded comorbidity variables for the analysis dataset
CHFhx <- rhc$chfhx
CARDIOhx <- rhc$cardiohx
DEMENThx <- rhc$dementhx
PSYCHhx <- rhc$psychhx
CHRPULhx <- rhc$chrpulhx
RENALhx <- rhc$renalhx
LIVERhx <- rhc$liverhx
GIBLEDhx <- rhc$gibledhx
MALIGHhx <- rhc$malighx
IMMUNhx <- rhc$immunhx
TRANSHhx <- rhc$transhx
AMIhx <- rhc$amihx

### Disease Severity and Clinical Characteristics

## Indicator of cancer diagnosis
Cancer <- rhc$ca

## Primary disease category
Category <- droplevels(rhc$cat1)

## Baseline disease severity score
Severity <- as.numeric(rhc$das2d3pc)

```

Construct Analysis Dataset

```
## Combine the outcome, treatment, and baseline covariates into a single  
## analysis-ready dataset  
data.set <- data.frame(  
  time = time,  
  Delta = Delta,  
  A = A,  
  
  Gender = Gender,  
  Age = Age,  
  Weight = Weight,  
  Race = Race,  
  Education = Education,  
  Income = Income,  
  InsuranceClass = InsuranceClass,  
  
  CHFhx = CHFhx,  
  CARDIOhx = CARDIOhx,  
  DEMENThx = DEMENThx,  
  PSYCHhx = PSYCHhx,  
  CHRPULhx = CHRPULhx,  
  RENALhx = RENALhx,  
  LIVERhx = LIVERhx,  
  GIBLEDhx = GIBLEDhx,  
  MALIGHx = MALIGHx,  
  IMMUNhx = IMMUNhx,  
  TRANShx = TRANShx,  
  AMIhx = AMIhx,  
  
  Cancer = Cancer,  
  Category = Category,  
  Severity = Severity  
)
```

Plotting Parameters

```
## Define colours used to distinguish the marginal structural Cox model  
## and the case-base model in survival and hazard ratio plots  
col.cox <- "red"  
col.cb <- "blue"  
  
## Define semi-transparent fill colours for confidence interval bands  
fill.cox <- rgb(1,0,0,0.20)  
fill.cb <- rgb(0,0,1,0.20)  
  
## Define semi-transparent colours for the exposed and control treatment groups  
col.exposed <- rgb(0,0,1,0.5)  
col.control <- rgb(1,0,0,0.5)
```

Propensity Score Estimation

```
## Set seed to ensure reproducibility of any downstream procedures involving
## randomization or resampling
set.seed(s)

## Estimate the propensity score, defined as the probability of receiving
## treatment conditional on the observed baseline covariates
propensity.score <- fitted(
  glm(
    A ~ Gender +
      Age +
      Weight +
      Race +
      Education +
      Income +
      InsuranceClass +
      CHFhx +
      CARDIOhx +
      DEMENThx +
      PSYCHhx +
      CHRPULhx +
      RENALhx +
      LIVERhx +
      GIBLEDhx +
      MALIGHx +
      IMMUNhx +
      TRANShx +
      AMIhx +
      Cancer +
      Category +
      Severity,
    data = data.set,
    family = binomial(link = "logit")
  )
)

## Separate estimated propensity scores by treatment group to assess the
## degree of overlap in treatment assignment probabilities
propensity.score.0 <- propensity.score[A == 0]
propensity.score.1 <- propensity.score[A == 1]
```

Propensity Score Overlap

```
## Plot the propensity score distributions for treated and untreated patients
## to visually assess the positivity/overlap condition
par(
  mar=c(2,2,2,0)
)
```

```

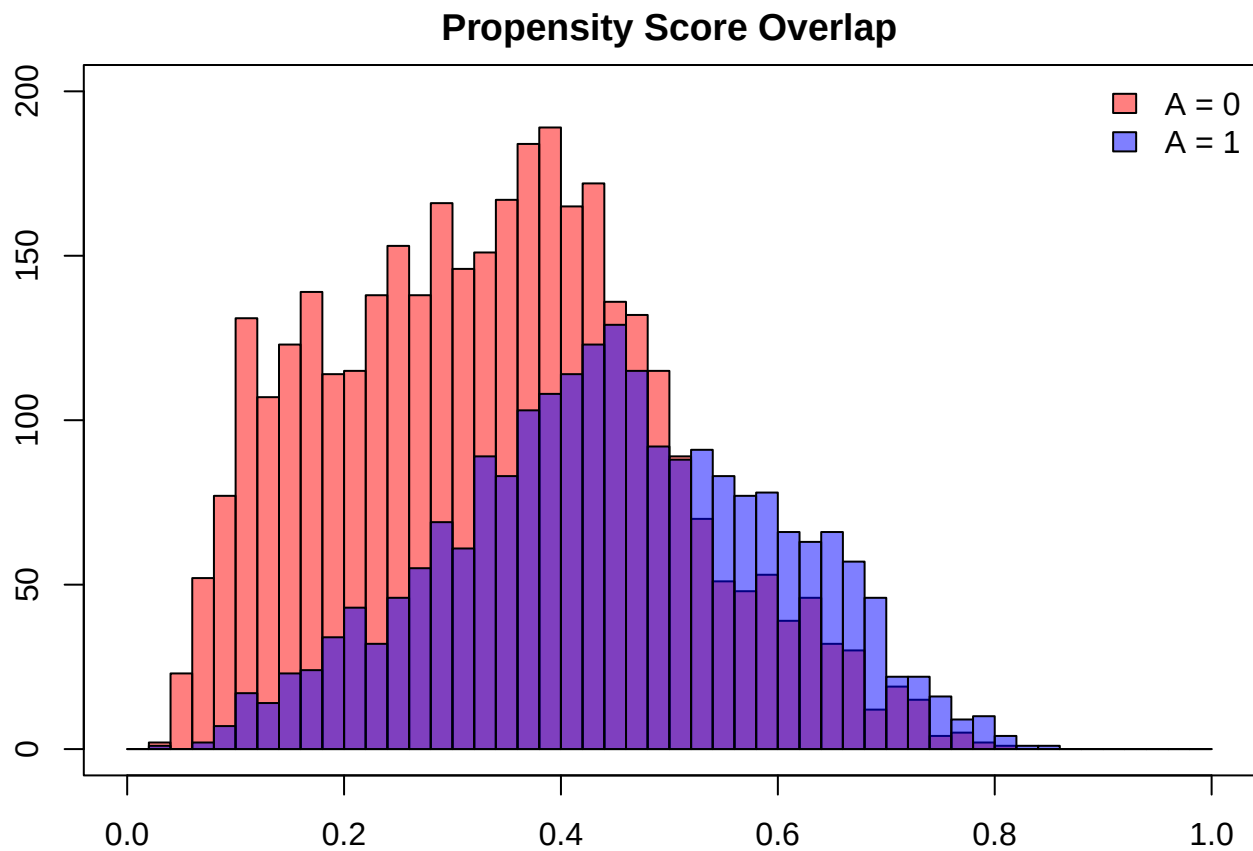
hist(
  propensity.score.0,
  col=col.control,
  breaks=seq(0,1,by=0.02),
  ylim=c(0,200),
  main="Propensity Score Overlap",
  xlab="Propensity Scores"
)

hist(
  propensity.score.1,
  col=col.exposed,
  breaks=seq(0,1,by=0.02),
  add=T
)

box()

legend(
  "topright",
  legend=c('A = 0', 'A = 1'),
  fill=c(col.control,col.exposed),
  border="black",
  bty="n"
)

```



Stabilized Inverse Probability Weights

```
## Estimate the marginal probability of treatment, used to construct the
## numerator of the stabilized treatment weights
p.A1 <- mean(A)

## Calculate unstabilized inverse probability of treatment weights (IPWs)
## by taking the inverse of the estimated probability of the treatment
## actually received
ipw.unstabilized <- ifelse(
  A == 1,
  1 / propensity.score,
  1 / (1 - propensity.score)
)

## Calculate stabilized IPWs by incorporating the marginal probability of
## treatment in the numerator. Stabilization reduces the variability of the
## weights while preserving their role in creating a weighted pseudo-population
ipw.stabilized <- ifelse(
  A == 1,
  p.A1 / propensity.score,
  (1 - p.A1) / (1 - propensity.score)
)
```

IPW Distribution

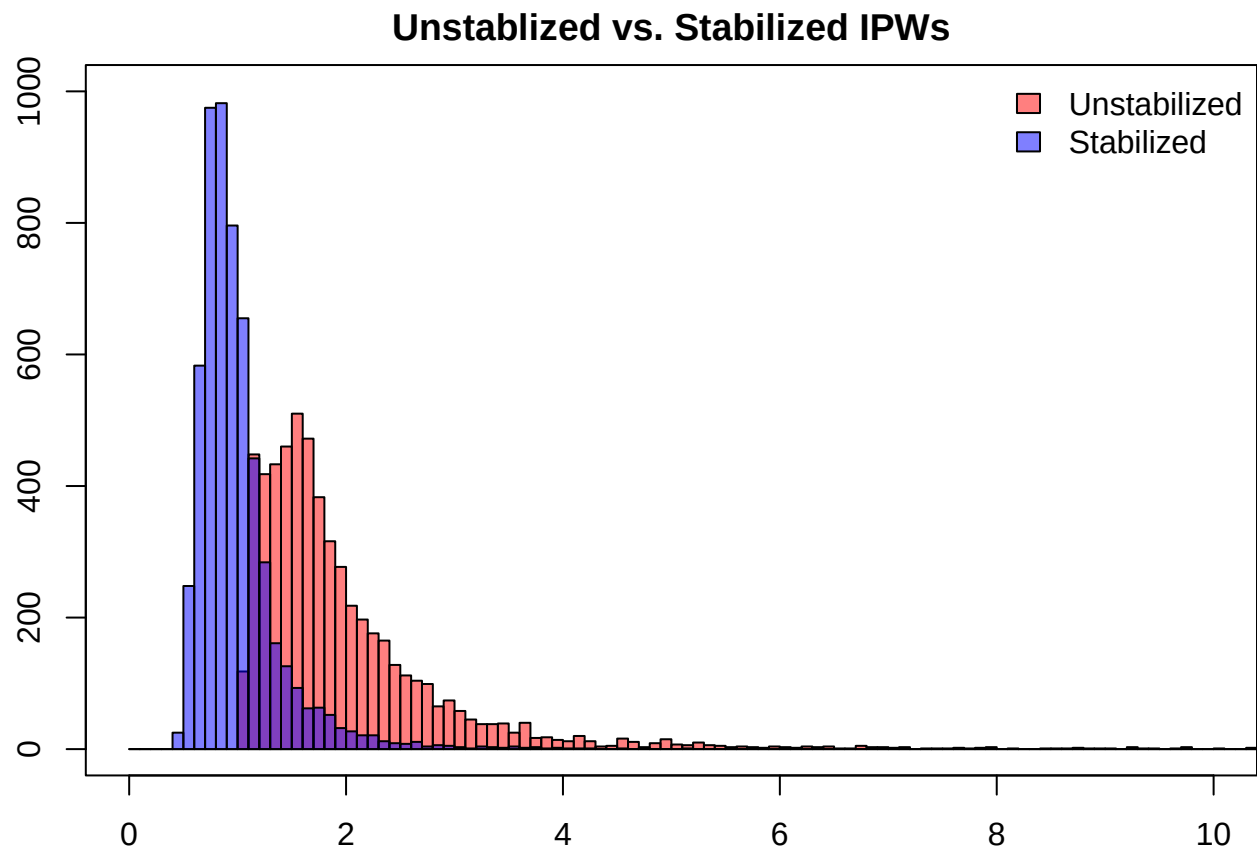
```
## Compare the distributions of the unstabilized and stabilized IPWs to
## assess the magnitude and variability of the resulting weights
par(
  mar=c(2,2,2,0)
)

hist(
  ipw.unstabilized,
  col=col.control,
  breaks=seq(0, max(ipw.unstabilized) + 0.5,by=0.1),
  ylim=c(0,1000),
  xlim=c(0, 10),
  main="Unstabilized vs. Stabilized IPWs",
  ylab="Frequency",
  xlab="IPWs"
)

hist(
  ipw.stabilized,
  col=col.exposed,
  breaks=seq(0, max(ipw.stabilized) + 0.5,by=0.1),
  add=T
)
```

```
box()

legend(
  "topright",
  legend=c('Unstabilized', 'Stabilized'),
  fill=c(col.control,col.exposed),
  border="black", bty="n"
)
```



Descriptive Statistics: Table One

```
## Specify the baseline covariates to be included in the descriptive table
covariates = c(
  "Gender",
  "Age",
  "Weight",
  "Race",
  "Education",
  "Income",
  "InsuranceClass",

  "CHFhx",
  "CARDIOhx",
  "DEMENThx",
  "PSYCHhx",
  "CHRPULhx",
  "RENALhx",
  "LIVERhx",
  "GIBLEDhx",
  "MALIGHx",
  "IMMUNhx",
  "TRANShx",
  "AMIhx",

  "Cancer",
  "Category",
  "Severity"
)

## Generate the unweighted baseline characteristics table, stratified by
## treatment group. Statistical significance tests are omitted because the
## primary objective is descriptive comparison of baseline characteristics.
table.one.original <- CreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set,
  test = FALSE
)

## Display standardized mean differences (SMDs) to quantify baseline
## differences between the treatment groups
print(table.one.original, smd = TRUE)
```

##	Stratified by A			
##	0	1		SMD
## n	3551	2184		
## Gender = Male (%)	1914 (53.9)	1278 (58.5)		0.093
## Age (mean (SD))	61.76 (17.29)	60.75 (15.63)		0.061
## Weight (mean (SD))	65.04 (29.50)	72.36 (27.73)		0.256
## Race (%)				0.036
## white	2753 (77.5)	1707 (78.2)		

##	black	585 (16.5)	335 (15.3)	
##	other	213 (6.0)	142 (6.5)	
##	Education (mean (SD))	11.57 (3.13)	11.86 (3.16)	0.091
##	Income (%)			0.142
##	Under \$11k	2081 (58.6)	1145 (52.4)	
##	\$11-\$25k	713 (20.1)	452 (20.7)	
##	\$25-\$50k	500 (14.1)	393 (18.0)	
##	> \$50k	257 (7.2)	194 (8.9)	
##	InsuranceClass (%)			0.194
##	No insurance	186 (5.2)	136 (6.2)	
##	Private	967 (27.2)	731 (33.5)	
##	Medicare	947 (26.7)	511 (23.4)	
##	Medicaid	454 (12.8)	193 (8.8)	
##	Private & Medicare	746 (21.0)	490 (22.4)	
##	Medicare & Medicaid	251 (7.1)	123 (5.6)	
##	CHFhx = Yes (%)	596 (16.8)	425 (19.5)	0.070
##	CARDIOhx = Yes (%)	567 (16.0)	446 (20.4)	0.116
##	DEMENThx = Yes (%)	413 (11.6)	151 (6.9)	0.163
##	PSYCHhx = Yes (%)	286 (8.1)	100 (4.6)	0.143
##	CHRPULhx = Yes (%)	774 (21.8)	315 (14.4)	0.192
##	RENALhx = Yes (%)	149 (4.2)	106 (4.9)	0.032
##	LIVERhx = Yes (%)	265 (7.5)	136 (6.2)	0.049
##	GIBLEDhx = Yes (%)	131 (3.7)	54 (2.5)	0.070
##	MALIGHx = Yes (%)	872 (24.6)	444 (20.3)	0.101
##	IMMUNhx = Yes (%)	907 (25.5)	636 (29.1)	0.080
##	TRANShx = Yes (%)	335 (9.4)	327 (15.0)	0.170
##	AMIHx = Yes (%)	105 (3.0)	95 (4.3)	0.074
##	Cancer (%)			0.107
##	No	2652 (74.7)	1727 (79.1)	
##	Yes	638 (18.0)	334 (15.3)	
##	Metastatic	261 (7.4)	123 (5.6)	
##	Category (%)			0.583
##	ARF	1581 (44.5)	909 (41.6)	
##	COPD	399 (11.2)	58 (2.7)	
##	CHF	247 (7.0)	209 (9.6)	
##	Cirrhosis	175 (4.9)	49 (2.2)	
##	Coma	341 (9.6)	95 (4.3)	
##	Colon Cancer	6 (0.2)	1 (0.0)	
##	Lung Cancer	34 (1.0)	5 (0.2)	
##	MOSF w/Malignancy	241 (6.8)	158 (7.2)	
##	MOSF w/Sepsis	527 (14.8)	700 (32.1)	
##	Severity (mean (SD))	20.37 (5.48)	20.70 (5.03)	0.063

Weighted Descriptive Statistics: Table Two

```
## Construct a survey design object using the stabilized IPWs. Setting ids = ~0
## specifies that observations are treated as independent sampling units.
data.set.ipw <- svydesign(
  ids = ~ 0,
  data = data.set,
  weights = ipw.stabilized
)

## Generate the weighted baseline characteristics table using the IPW-defined
## pseudo-population, stratified by treatment group
table.one.ipw <- svyCreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set.ipw,
  test = FALSE
)

## Display standardized mean differences for the weighted covariate
## distributions to assess residual imbalance after IPW
print(table.one.ipw, smd = TRUE)
```

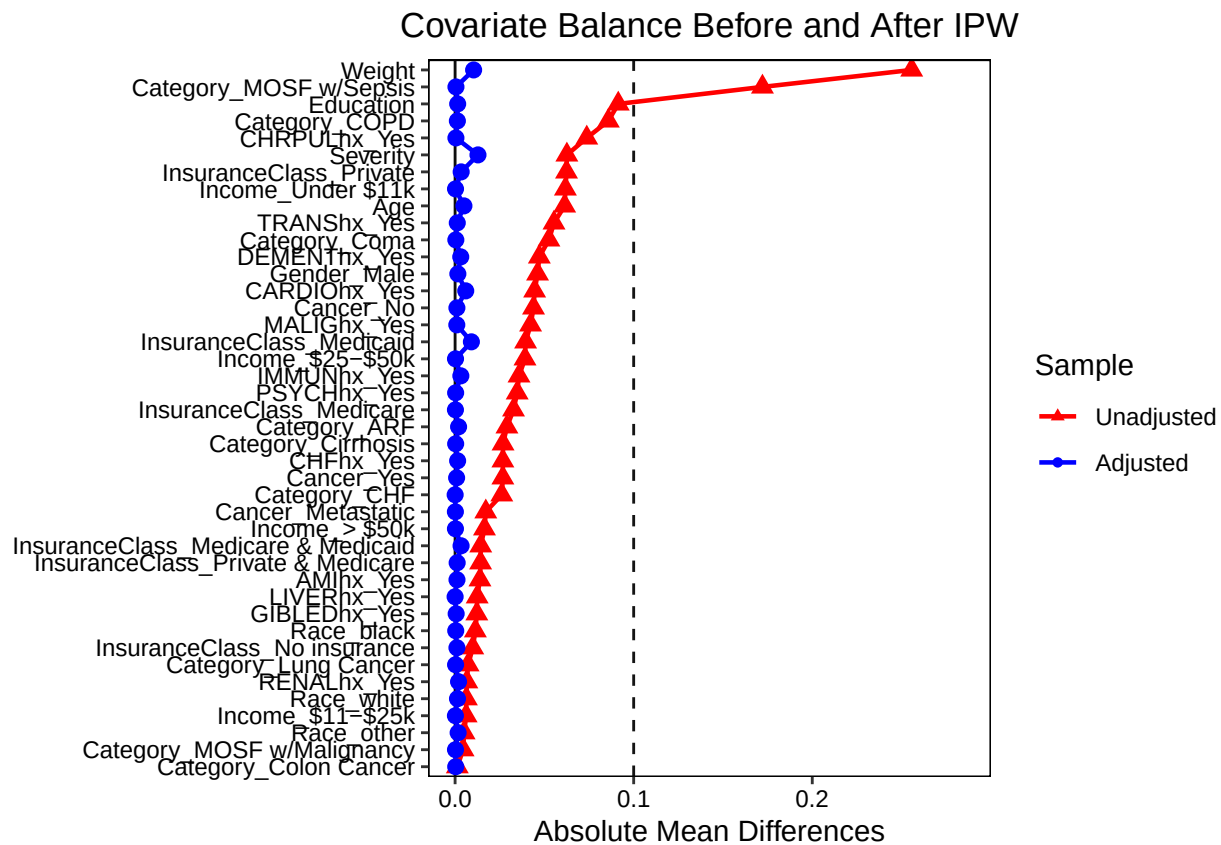
##	Stratified by A			
##	0	1		SMD
## n	3554.6	2178.4		
## Gender = Male (%)	1968.2 (55.4)	1209.5 (55.5)		0.003
## Age (mean (SD))	61.33 (17.24)	61.25 (15.74)		0.005
## Weight (mean (SD))	67.93 (29.11)	68.23 (28.44)		0.010
## Race (%)				0.007
## white	2759.7 (77.6)	1694.1 (77.8)		
## black	576.6 (16.2)	354.1 (16.3)		
## other	218.3 (6.1)	130.1 (6.0)		
## Education (mean (SD))	11.69 (3.14)	11.69 (3.08)		0.001
## Income (%)				0.001
## Under \$11k	1997.5 (56.2)	1224.6 (56.2)		
## \$11-\$25k	723.4 (20.4)	443.7 (20.4)		
## \$25-\$50k	555.6 (15.6)	340.0 (15.6)		
## > \$50k	278.1 (7.8)	170.0 (7.8)		
## InsuranceClass (%)				0.031
## No insurance	198.0 (5.6)	119.1 (5.5)		
## Private	1047.8 (29.5)	634.7 (29.1)		
## Medicare	899.1 (25.3)	550.6 (25.3)		
## Medicaid	404.3 (11.4)	267.6 (12.3)		
## Private & Medicare	775.2 (21.8)	472.6 (21.7)		
## Medicare & Medicaid	230.1 (6.5)	133.7 (6.1)		
## CHFhx = Yes (%)	634.9 (17.9)	392.2 (18.0)		0.004
## CARDIOhx = Yes (%)	636.8 (17.9)	403.2 (18.5)		0.015
## DEMENThx = Yes (%)	347.5 (9.8)	206.1 (9.5)		0.011
## PSYCHhx = Yes (%)	237.3 (6.7)	146.1 (6.7)		0.001
## CHRPULhx = Yes (%)	669.0 (18.8)	411.0 (18.9)		0.001
## RENALhx = Yes (%)	161.7 (4.5)	103.3 (4.7)		0.009

##	LIVERhx = Yes (%)	250.2 (7.0)	153.3 (7.0)	<0.001
##	GIBLEDhx = Yes (%)	114.7 (3.2)	69.0 (3.2)	0.003
##	MALIGHx = Yes (%)	817.4 (23.0)	503.0 (23.1)	0.002
##	IMMUNhx = Yes (%)	965.2 (27.2)	598.5 (27.5)	0.007
##	TRANShx = Yes (%)	412.5 (11.6)	255.5 (11.7)	0.004
##	AMIHx = Yes (%)	122.4 (3.4)	77.3 (3.5)	0.006
##	Cancer (%)			0.002
##	No	2712.1 (76.3)	1659.9 (76.2)	
##	Yes	603.9 (17.0)	371.9 (17.1)	
##	Metastatic	238.5 (6.7)	146.5 (6.7)	
##	Category (%)			0.011
##	ARF	1541.0 (43.4)	948.6 (43.5)	
##	COPD	282.7 (8.0)	170.5 (7.8)	
##	CHF	282.7 (8.0)	173.3 (8.0)	
##	Cirrhosis	138.6 (3.9)	85.6 (3.9)	
##	Coma	270.2 (7.6)	164.7 (7.6)	
##	Colon Cancer	4.4 (0.1)	3.4 (0.2)	
##	Lung Cancer	24.1 (0.7)	14.2 (0.7)	
##	MOSF w/Malignancy	246.5 (6.9)	150.6 (6.9)	
##	MOSF w/Sepsis	764.4 (21.5)	467.4 (21.5)	
##	Severity (mean (SD))	20.48 (5.53)	20.55 (4.95)	0.013

Covariate Balance Before and After IPW

```
## Assess covariate balance before and after application of the stabilized
## inverse probability weights. The unweighted and weighted SMDs are retained
## to facilitate direct comparison of balance across treatment groups.
balance <- bal.tab(
  A ~ Gender +
    Age +
    Weight +
    Race +
    Education +
    Income +
    InsuranceClass +
    CHFhx +
    CARDIOhx +
    DEMENThx +
    PSYCHhx +
    CHRPULhx +
    RENALhx +
    LIVERhx +
    GIBLEDhx +
    MALIGHx +
    IMMUNhx +
    TRANShx +
    AMIhx +
    Cancer +
    Category +
    Severity,
  data = data.set,
  weights = ipw.stabilized,
  method = "weighting",
  estimand = "Average Treatment Effect",
  un = TRUE
)

## Visualize the standardized mean differences before and after IPW.
## An absolute SMD below 0.1 is used as the criterion for acceptable
## covariate balance.
love.plot(
  balance,
  stats = "mean.diffs",
  threshold = 0.1,
  abs = TRUE,
  line = TRUE,
  var.order = "unadjusted",
  colors = c(col.control, col.exposed),
  shapes = c(17, 16),
  limits = c(m = c(0, 0.3)),
  title = "Covariate Balance Before and After IPW",
  labels = TRUE
)
```



```
## Covariate balance improved substantially following application of the
## stabilized inverse probability weights, indicating improved comparability
## between treatment groups in the weighted pseudo-population.
```

Estimation

Assumptions

Non-Informative Censoring

Modified Case-Base Smooth Hazard Model

```
## Modified implementation of fitSmoothHazard() from the casebase package.
## The function extends the standard case-base fitting procedure to allow
## observation-level weights, including inverse probability weights (IPWs).
FSH <- function(
  formula,
  data,
  time,
  w = rep(1, nrow(data)),
  ratio = 100, ...
) {

  ## Preserve the original function call for reproducibility and method
  ## dispatch in downstream model summaries
  cl <- match.call()

  ## Explicitly expand "." in the model formula so that all covariates in the
  ## supplied dataset are represented in the model specification
  formula <- expand.dot.formula(
    formula,
    data = data
  )

  ## Extract the event variable from the left-hand side of the model formula
  eventVar <- all.vars(formula[[2]])

  ## If the follow-up time variable is not explicitly supplied, infer it from
  ## the data using the event variable and the standard case-base structure
  if (missing(time)) {
    varNames <- checkArgsTimeEvent(
      data = data,
      event = eventVar
    )

    timeVar <- varNames$time
  } else {
    ## Otherwise, use the time variable explicitly supplied by the user
    timeVar <- time
  }

  ## Identify the distinct event types present in the data to determine whether
  ## the analysis involves a single event or competing risks
  typeEvents <- sort(unique(data[[eventVar]]))
}
```

```

# Case-Base Sampling

## Generate the case-base sample when the input data have not already been
## sampled. The original dataset is retained so that the supplied observation
## weights can be incorporated into the case-base sampling procedure.
if (!inherits(data, "cbData")) {

  originalData <- as.data.frame(data)
  originalData$w <- w

  ## Sample cases and controls from the person-time data. The sampling ratio
  ## controls the number of person-moments sampled relative to observed cases.
  sampleData <- sampleCaseBase(
    originalData,
    timeVar,
    eventVar,
    comprisk = (length(typeEvents) > 2),
    ratio
  )
} else {
  ## If the input is already a case-base dataset, retain it directly.
  ## The original unsampled data are unavailable at this stage.
  originalData <- data.frame()
  sampleData <- data
}

# Weighted Case-Base Model

## Include the case-base sampling offset in the model formula. This offset
## accounts for the sampling mechanism when estimating the hazard function.
formula <- update(formula, ~ . + offset(offset))

## Fit a weighted binomial regression model to the case-base sample.
## Observation-level weights allow the model to incorporate IPWs in addition
## to the case-base sampling structure.
fittingFunction <- function(formula) glm(
  formula,
  data = sampleData,
  family = binomial,
  weights = w
)

## Estimate the smooth hazard model using the weighted case-base sample
out <- fittingFunction(formula)

# Model Object and Metadata

## Preserve the lower-level glm call while replacing the displayed call with
## the original FSH call for consistency with the custom model interface

```

```

out$lower_call <- out$call
out$call <- cl

## Store the original data and key variable names for downstream methods
## such as prediction and plotting
out$originalData <- originalData
out$typeEvents <- typeEvents
out$timeVar <- timeVar
out$eventVar <- eventVar

## Record the number of case and background person-moments in the sampled
## case-base dataset
num_pm <- table(sampleData[[eventVar]])
out$num_cm <- num_pm[2]
out$num_bm <- num_pm[1]
out$formula <- formula

## Store the case-base sampling offset separately, then reset the model's
## offset to zero so that absolute risk estimation can be performed without
## retaining the sampling correction in the prediction stage
out$offset <- out$data$offset
out$data$offset <- 0

## Assign a custom class so that methods defined for single-event case-base
## models can be applied to the fitted object
class(out) <- c("singleEventCB", class(out))

return(out)
}

```

Formula Expansion Helper

```

## Expand "." in a model formula into the complete set of covariates available
## in the supplied dataset. This ensures that the modified FSH function can
## explicitly evaluate formulas containing "." before fitting the model.
expand.dot.formula <- function(formula, data = NULL) {

  ## Check whether the formula contains the "." shorthand for all available
  ## predictors
  if (isTRUE("." %in% all.vars(formula))) {

    ## Preserve the original formula attributes while expanding the terms
    att <- attributes(formula)

    try.terms <- try(
      stats::terms(formula, data = data),
      silent = TRUE
    )

    ## Replace the shorthand formula with its explicitly expanded version when
    ## the terms object can be constructed successfully

```

```
if (!is(try.terms, "try-error")) {  
  formula <- formula(try.terms)  
}  
  
## Restore the original formula attributes after expansion  
attributes(formula) <- att  
}  
  
## Return the expanded formula  
return(formula)  
}
```

Case-Base Model Setup

```
## Set the case-base sampling ratio used across all case-base model fits.  
## A ratio of 100 samples 100 background person-moments per observed case.  
fsh.ratio <- 100
```

Counterfactual Survival Curve Setup

```
## Define the two treatment levels used to generate counterfactual survival  
## curves: A = 0 represents no RHC and A = 1 represents RHC  
trt.types <- c(0, 1)  
  
## Create a single-treatment profile for prediction under the treated condition  
individual.profile.data <- data.frame(A = 1)  
  
## Create a prediction dataset containing both treatment levels for estimation  
## of counterfactual survival under each treatment condition  
survival.curves.data <- data.frame(A = trt.types)  
  
# Time Grid Definition  
  
## Determine the maximum observed follow-up time in the analysis dataset  
tau.max <- max(data.set$time, na.rm = TRUE)  
  
## Restrict the primary survival-curve display to the first 150 days  
## (approximately four months) to focus on the relevant follow-up period  
t.max <- 150  
  
## Define the time grid used for RMST integration, spanning the full observed  
## follow-up period at one-day intervals  
tau.grid <- seq(0, tau.max, by = 1)  
  
## Define the time grid used for plotting and estimating survival curves over  
## a restricted four-month follow-up period  
t.grid <- seq(0, t.max, by = 1)
```

Model 1: Unadjusted Marginal Structural Cox Model

```
## Set seed to ensure reproducibility of any stochastic procedures used
## during model fitting
set.seed(s)

## Fit an unadjusted Cox proportional hazards model with treatment as the
## sole predictor. The resulting coefficient estimates the marginal
## log-hazard ratio comparing RHC with no RHC.
model.1 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  robust = TRUE
)

## Extract the treatment coefficient and its standard error
beta.1 <- coef(model.1)["A"]
se.1 <- sqrt(diag(vcov(model.1))["A"])

## Transform the treatment coefficient from the log-hazard ratio scale to the
## hazard ratio scale and calculate the corresponding 95% Wald confidence
## interval
cat("HR-hat:", exp(beta.1), "\n")
```

```
## HR-hat: 1.164139
```

```
cat("SE of HR-hat:", se.1, "\n")
```

```
## SE of HR-hat: 0.0342025
```

```
cat("95% CI: (", exp(beta.1 + c(-1,1)*1.96*se.1), ")\n")
```

```
## 95% CI: ( 1.088657 1.244854 )
```

Model 2: Unadjusted Case-Base Model

```
## Set seed to ensure reproducibility of the case-base sampling procedure
set.seed(s)

## Fit an unadjusted smooth hazard model using case-base sampling, with
## treatment as the sole predictor. The sampling ratio determines the number
## of background person-moments sampled for each observed case.
model.2 <- fitSmoothHazard(
  Delta ~ A,
  data = data.set,
  ratio = fsh.ratio
)
```

```
## Extract the treatment coefficient and calculate its robust standard error
## using the heteroskedasticity-consistent covariance estimator
beta.2 <- coef(model.2)["A"]
se.2 <- sqrt(diag(vcovHC(model.2))["A"])

## Transform the treatment coefficient to the hazard ratio scale and calculate
## the corresponding 95% Wald confidence interval
cat("HR-hat:", exp(beta.2), "\n")
```

```
## HR-hat: 1.205197
```

```
cat("SE of HR-hat:", se.2, "\n")
```

```
## SE of HR-hat: 0.03364396
```

```
cat("95% CI: (", exp(beta.2 + c(-1,1)*1.96*se.2), ")\n")
```

```
## 95% CI: ( 1.128288 1.28735 )
```

Model 3: Covariate-Adjusted Cox Model

```
## Set seed to ensure reproducibility of any stochastic procedures used
## during model fitting
set.seed(s)

## Fit a covariate-adjusted Cox proportional hazards model. Treatment is
## adjusted for the full set of observed baseline demographic, socioeconomic,
## comorbidity, and disease-severity covariates.
model.3 <- coxph(
  Surv(time, Delta) ~ A +
    Gender +
    Age +
    Weight +
    Race +
    Education +
    Income +
    InsuranceClass +
    CHFhx +
    CARDIOhx +
    DEMENThx +
    PSYCHhx +
    CHRPULhx +
    RENALhx +
    LIVERhx +
    GIBLEDhx +
    MALIGHx +
    IMMUNhx +
    TRANShx +
    AMIhx +
    Cancer +
    Category +
    Severity,
  data = data.set,
  robust = TRUE
)

## Extract the treatment coefficient and its robust standard error
beta.3 <- coef(model.3)["A"]
se.3 <- sqrt(diag(vcov(model.3))["A"])

## Transform the treatment coefficient to the hazard ratio scale and calculate
## the corresponding 95% Wald confidence interval
cat("HR-hat:", exp(beta.3), "\n")

## HR-hat: 1.288328

cat("SE of HR-hat:", se.3, "\n")

## SE of HR-hat: 0.03765539
```

```
cat("95% CI: (", exp(beta.3 + c(-1,1)*1.96*se.3), ")\n")
```

```
## 95% CI: ( 1.196667 1.387009 )
```

Model 4: Covariate-Adjusted Case-Base Model

```
## Set seed to ensure reproducibility of the case-base sampling procedure
set.seed(s)

## Fit a covariate-adjusted smooth hazard model using case-base sampling.
## The model uses the same treatment and baseline covariates as the adjusted
## Cox model to provide a directly comparable alternative estimator.
model.4 <- fitSmoothHazard(
  Delta ~ A +
    Gender +
    Age +
    Weight +
    Race +
    Education +
    Income +
    InsuranceClass +
    CHFhx +
    CARDIOhx +
    DEMENThx +
    PSYCHhx +
    CHRPULhx +
    RENALhx +
    LIVERhx +
    GIBLEDhx +
    MALIGHx +
    IMMUNhx +
    TRANShx +
    AMIhx +
    Cancer +
    Category +
    Severity,
  data = data.set,
  ratio = fsh.ratio
)

## Extract the treatment coefficient and calculate its heteroskedasticity-
## consistent standard error
beta.4 <- coef(model.4)["A"]
se.4 <- sqrt(diag(vcovHC(model.4))["A"])

## Transform the treatment coefficient to the hazard ratio scale and calculate
## the corresponding 95% Wald confidence interval
cat("HR-hat:", exp(beta.4), "\n")

## HR-hat: 1.371035

cat("SE of HR-hat:", se.4, "\n")

## SE of HR-hat: 0.036884
```

```
cat("95% CI: (", exp(beta.4 + c(-1,1)*1.96*se.4), ")\n")
```

```
## 95% CI: ( 1.275417 1.473821 )
```

Model 5: Marginal Structural Cox Model

```
## Set seed to ensure reproducibility of any stochastic procedures used
## during model fitting
set.seed(s)

## Fit the marginal structural Cox model using stabilized inverse probability
## weights (IPWs). The treatment-only model estimates the marginal treatment
## effect in the IPW-defined pseudo-population.
model.5 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  weights = ipw.stabilized,
  robust = TRUE
)

## Extract the treatment coefficient and its robust standard error
beta.5 <- coef(model.5)["A"]
se.5 <- sqrt(diag(vcov(model.5))["A"])

## Transform the treatment coefficient to the hazard ratio scale and calculate
## the corresponding 95% Wald confidence interval
cat("HR-hat:", exp(beta.5), "\n")

## HR-hat: 1.232318

cat("SE of HR-hat:", se.5, "\n")

## SE of HR-hat: 0.03764234

cat("95% CI: (", exp(beta.5 + c(-1,1)*1.96*se.5), ")\n")

## 95% CI: ( 1.144672 1.326675 )
```

Model 6: IPW Case-Base Proportional Hazards Model

```
## Set seed to ensure reproducibility of the case-base sampling procedure
set.seed(s)

## Fit the modified smooth hazard model using case-base sampling and stabilized
## IPWs. The custom FSH function incorporates the IPWs as observation-level
## weights while accounting for the case-base sampling mechanism.
model.6 <- FSH(
  Delta ~ A,
  data = data.set,
  ratio = fsh.ratio,
  w = ipw.stabilized
)

## Extract the treatment coefficient and calculate its heteroskedasticity-
## consistent standard error
beta.6 <- coef(model.6)["A"]
se.6 <- sqrt(diag(vcovHC(model.6))["A"])

## Transform the treatment coefficient to the hazard ratio scale and calculate
## the corresponding 95% Wald confidence interval
cat("HR-hat:", exp(beta.6), "\n")

## HR-hat: 1.274666

cat("SE of HR-hat:", se.6, "\n")

## SE of HR-hat: 0.03693444

cat("95% CI: (", exp(beta.6 + c(-1,1)*1.96*se.6), ")\n")

## 95% CI: ( 1.185652 1.370363 )
```

Survival Curve Prediction Setup

```
## Define the two treatment levels used to generate counterfactual survival  
## curves under no RHC (A = 0) and RHC (A = 1)  
trt.types <- c(0, 1)  
  
## Define the treatment profile used for individual-level prediction  
individual.profile.data <- data.frame(A = 1)  
  
## Create a prediction dataset containing both treatment levels for estimation  
## of the corresponding counterfactual survival curves  
survival.curves.data <- data.frame(A = trt.types)  
  
## Determine the maximum observed follow-up time and define the restricted  
## four-month follow-up period used for focused visualization  
tau.max <- max(data.set$time, na.rm = TRUE)  
t.max <- 150  
  
## Define the full follow-up grid and the restricted four-month grid  
tau.grid <- seq(0, tau.max, by = 1)  
t.grid <- seq(0, t.max, by = 1)
```

Model 7: Unadjusted Case-Base Model with Time-Varying Hazard Ratio

```
## Set seed to ensure reproducibility of the case-base sampling procedure
set.seed(s)

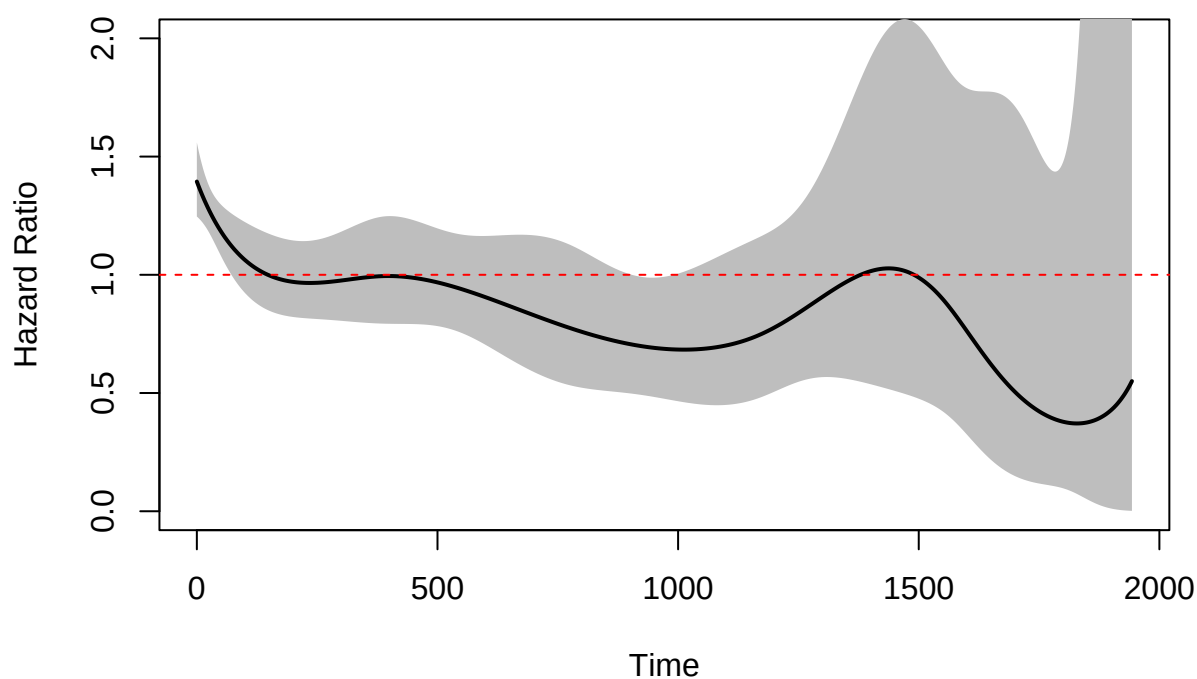
## Fit an unadjusted case-base smooth hazard model allowing the treatment
## effect to vary over time. The spline term models the baseline time effect,
## while its interaction with treatment permits a non-constant hazard ratio.
model.7 <- FSH(
  Delta ~ pspline(time, df = 2) * A,
  data = data.set,
  ratio = fsh.ratio
)
```

Time-Varying Hazard Ratio: Full Follow-Up

```
## Display the estimated treatment hazard ratio as a function of follow-up
## time, including its 95% confidence interval
plot(
  model.7,
  type = "hr",
  newdata = expand.grid(
    time = tau.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(0,2),
  xlab="Time",
  ylab="Hazard Ratio",
  main="Model 7: Variable Hazard Ratio"
)

## Add a reference line corresponding to a hazard ratio of 1, representing
## no instantaneous treatment effect
abline(h = 1, col = "red", lty = 2)
```

Model 7: Variable Hazard Ratio

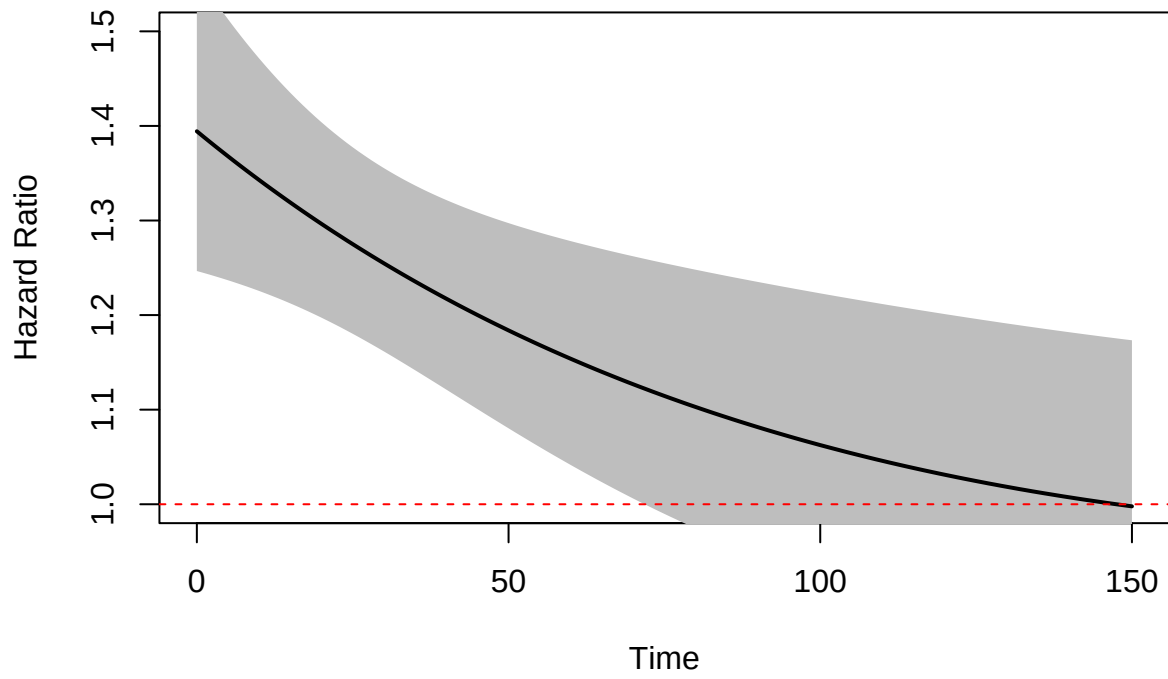


Time-Varying Hazard Ratio: First Four Months

```
## Restrict the hazard-ratio plot to the first 150 days (approximately four
## months) to provide a more detailed view of the early follow-up period
plot(
  model.7,
  type = "hr",
  newdata = expand.grid(
    time = t.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(1,1.5),
  xlab="Time",
  ylab="Hazard Ratio",
  main="Model 7: Variable Hazard Ratio - First 4 Months"
)

## Add the no-effect reference line
abline(h = 1, col = "red", lty = 2)
```

Model 7: Variable Hazard Ratio – First 4 Months



Model 8: IPW Case-Base Model with Time-Varying Hazard Ratio

```
## Set seed to ensure reproducibility of the case-base sampling procedure
set.seed(s)

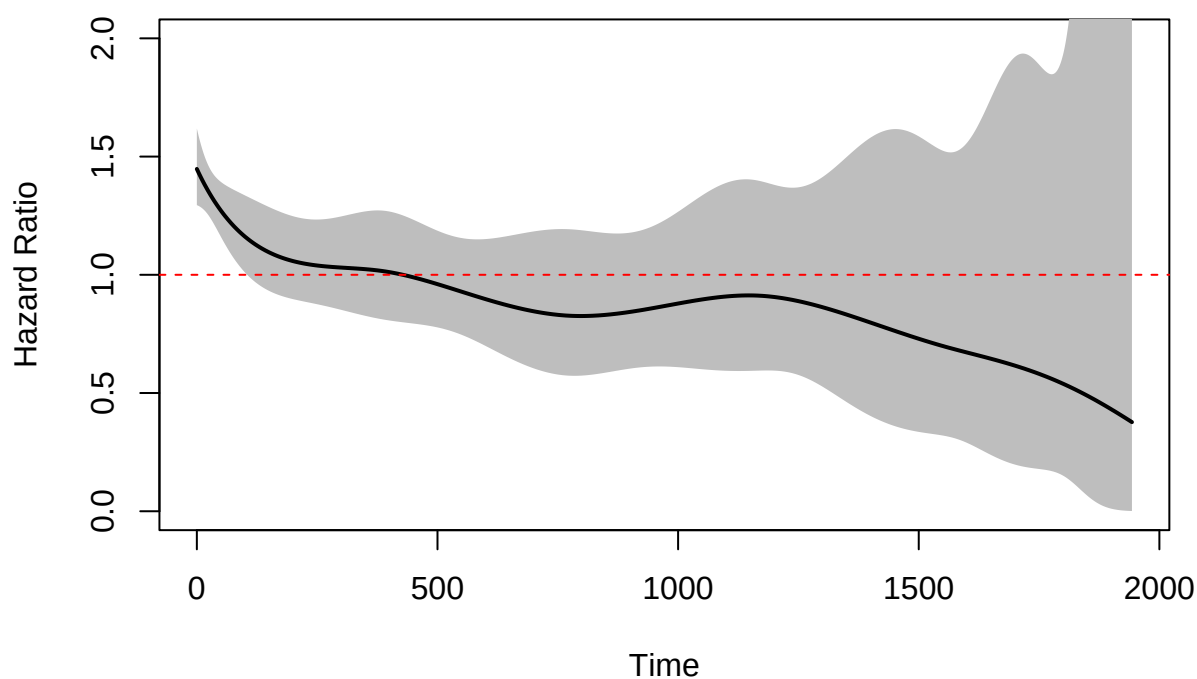
## Fit the IPW case-base smooth hazard model, allowing the treatment effect
## to vary over time. Stabilized IPWs are incorporated through the modified
## FSH function to estimate the treatment effect in the weighted
## pseudo-population.
model.8 <- FSH(
  Delta ~ pspline(time, df = 2) * A,
  data = data.set,
  ratio = fsh.ratio,
  w = ipw.stabilized
)
```

Time-Varying Hazard Ratio: Full Follow-Up

```
## Display the estimated time-varying treatment hazard ratio and its 95%
## confidence interval over the full follow-up period
plot(
  model.8,
  type = "hr",
  newdata = expand.grid(
    time = tau.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(0,2),
  xlab="Time",
  ylab="Hazard Ratio",
  main="Model 8: Variable Hazard Ratio"
)

## Add the no-effect reference line
abline(h = 1, col = "red", lty = 2)
```

Model 8: Variable Hazard Ratio

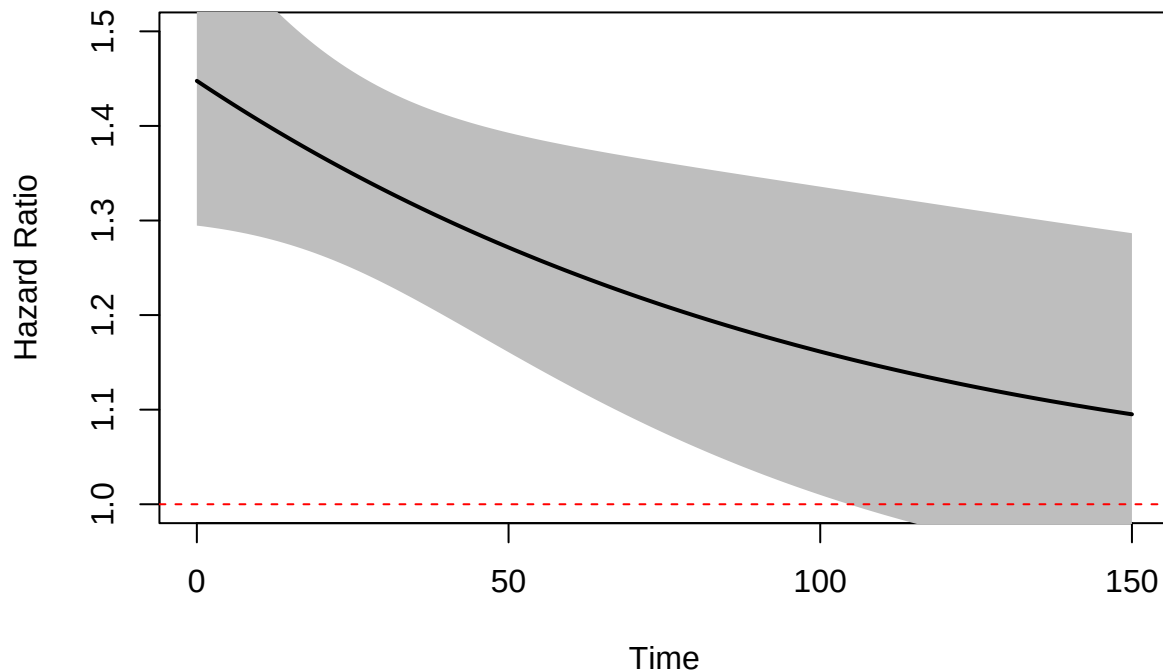


Time-Varying Hazard Ratio: First Four Months

```
## Restrict the hazard-ratio plot to the first 150 days to provide a more
## detailed visualization of the early treatment effect
plot(
  model.8,
  type = "hr",
  newdata = expand.grid(
    time = t.grid,
    A = trt.types
  ),
  var = "A",
  xvar = "time",
  ci = TRUE,
  ylim = c(1,1.5),
  xlab="Time",
  ylab="Hazard Ratio",
  main="Model 8: Variable Hazard Ratio - First 4 Months"
)

## Add the no-effect reference line
abline(h = 1, col = "red", lty = 2)
```

Model 8: Variable Hazard Ratio – First 4 Months



Head-to-Head Comparison: Models 7 and 8

```
## Extract the estimated time-varying treatment hazard ratios and their 95%
## confidence intervals from the unadjusted and IPW-adjusted case-base models.
## The resulting estimates are used to construct a common comparison plot.
```

```
get.hr <- function(model) {
  invisible(pdf(NULL))
```

```
  out <- plot(
    model,
    type = "hr",
    newdata = data.frame(
      time = tau.grid,
      A = 0
    ),
    var = "A",
    xvar = "time",
    ci = TRUE
  )
```

```
  invisible(dev.off())
```

```
  out
```

```

}

## Extract the time-varying hazard-ratio estimates from Models 7 and 8
hr.7 <- get.hr(model.7)
hr.8 <- get.hr(model.8)

# Combined Hazard-Ratio Plot

## Initialize an empty plotting region using the time and hazard-ratio ranges
## from the extracted estimates. The curves and confidence intervals for both
## models are added to this common plotting region below.
plot(
  hr.7$time,
  hr.7$hazard_ratio,
  type = "n",
  ylim = c(0, 2),
  xlab = "Time",
  ylab = "Hazard Ratio",
  main = "Head-to-head Comparison of Variable HR Models"
)

# 95% Confidence Intervals

## Add the 95% confidence interval band for the unadjusted case-base model
polygon(
  c(hr.7$time, rev(hr.7$time)),
  c(hr.7$lowerbound, rev(hr.7$upperbound)),
  col = adjustcolor("red", alpha.f = 0.20),
  border = NA
)

## Add the 95% confidence interval band for the IPW-adjusted case-base model
polygon(
  c(hr.8$time, rev(hr.8$time)),
  c(hr.8$lowerbound, rev(hr.8$upperbound)),
  col = adjustcolor("blue", alpha.f = 0.20),
  border = NA
)

# Estimated Hazard-Ratio Curves

## Overlay the estimated time-varying hazard-ratio curve from the unadjusted
## case-base model
lines(
  hr.7$time,
  hr.7$hazard_ratio,
  col = "red",
  lwd = 2
)

## Overlay the estimated time-varying hazard-ratio curve from the IPW-adjusted
## case-base model
lines(

```

```

hr.8$time,
hr.8$hazard_ratio,
col = "blue",
lwd = 2
)

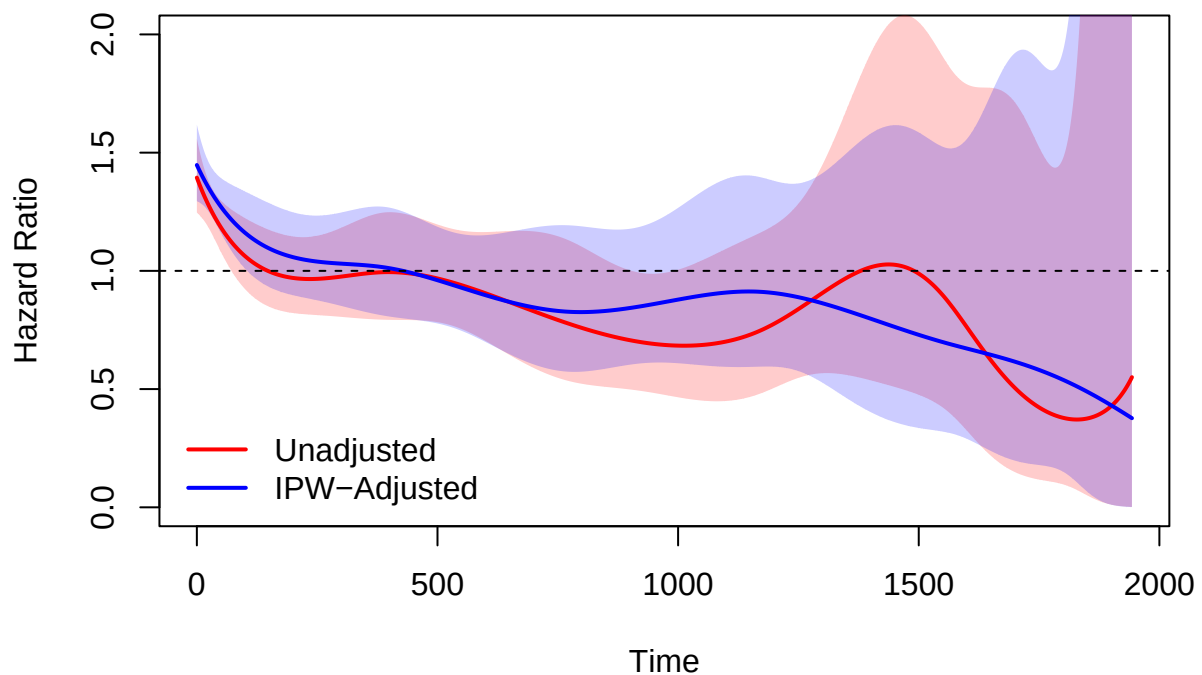
# Reference Line and Legend

## Add a horizontal reference line at HR = 1, corresponding to no difference
## in the instantaneous hazard between treatment groups
abline(
  h = 1,
  col = "black",
  lty = 2
)

## Identify the two estimation approaches shown in the combined plot
legend(
  "bottomleft",
  legend = c("Unadjusted", "IPW-Adjusted"),
  col = c("red", "blue"),
  lwd = 2,
  bty = "n"
)

```

Head-to-head Comparison of Variable HR Models



Restricted Mean Survival Time (RMST)

```
## Numerically integrate a survival curve over the specified follow-up period  
## using the trapezoidal rule. The cumulative integral at each time point  
## represents the RMST up to that point.  
rmst.fn <- function(time, surv) {  
  dt <- diff(time)  
  mids <- (head(surv, -1) + tail(surv, -1)) / 2  
  c(0, cumsum(mids * dt))  
}
```

MS-Cox RMST and Standard Error

```
## Calculate the counterfactual RMST contrast and its standard error from the
## marginal structural Cox model. The RMST contrast is defined as the
## difference in restricted mean survival between RHC and no RHC.
cox.rmst.list <- function(
  ms.cox, t.grid, cox.psi.se, survival.curves.data
) {

  # Counterfactual Survival Curves

  ## Estimate the counterfactual survival curves under both treatment levels
  ## from the fitted marginal structural Cox model
  sf <- survfit(
    ms.cox,
    newdata = survival.curves.data
  )

  ## Interpolate the counterfactual survival curve under no RHC onto the
  ## specified time grid
  S0 <- approx(
    sf$time,
    sf$surv[, 1],
    t.grid,
    method = "constant",
    rule = 2
  )$y

  ## Interpolate the counterfactual survival curve under RHC onto the
  ## specified time grid
  S1 <- approx(
    sf$time,
    sf$surv[, 2],
    t.grid,
    method = "constant",
    rule = 2
  )$y

  # RMST Contrast

  ## Calculate the RMST contrast as the difference between the restricted mean
  ## survival under RHC and no RHC
  cox.rmst <- rmst.fn(t.grid, S1) - rmst.fn(t.grid, S0)

  # Delta-Method Standard Error

  ## Construct the gradient of the RMST functional with respect to the
  ## treatment log-hazard coefficient. For the Cox model, the derivative
  ## involves the integral of  $S_1(t) \log\{S_1(t)\}$  over the restriction period.
  integrand <- ifelse(S1 > 0, S1 * log(S1), 0)
  g.cox <- rmst.fn(t.grid, integrand)

  ## Propagate uncertainty from the estimated Cox treatment coefficient to the
```

```
## RMST contrast using the delta method
cox.rmst.se <- abs(g.cox) * cox.psi.se

return(list(
  cox.rmst = cox.rmst,
  cox.rmst.se = cox.rmst.se
))
}
```

IPW Case-Base RMST and Standard Error

```
## Calculate the RMST contrast and its standard error from the IPW case-base
## model. The survival curves are reconstructed from the fitted model and the
## variance of the RMST contrast is obtained using a numerical delta method.
cb.rmst.list <- function(cb.ipw, cb.var, t.grid, constant.HR) {

  ## Extract the estimated model coefficients and the corresponding model
  ## terms used to construct the design matrix
  beta.hat <- coef(cb.ipw)
  term.form <- delete.response(terms(cb.ipw))

  # Counterfactual Survival Function

  ## Construct the counterfactual survival function under treatment level a
  ## directly from the case-base model's design matrix
  cb.surv <- function(beta, a, grid) {

    ## Create prediction data over the specified time grid under treatment
    ## level a
    nd <- data.frame(
      time = grid,
      A = a,
      offset = 0
    )

    ## Construct the model matrix using the same terms as the fitted model and
    ## align its columns with the estimated coefficient vector
    X <- model.matrix(term.form, data = nd)
    X <- X[, names(beta), drop = FALSE]

    ## Convert the linear predictor to the estimated hazard function
    lambda <- exp(as.vector(X %*% beta))

    ## Numerically integrate the estimated hazard using the trapezoidal rule
    ## to obtain the cumulative hazard
    Lambda <- c(
      0,
      cumsum(
        diff(grid) * (head(lambda, -1) + tail(lambda, -1)) / 2
      )
    )

    ## Convert cumulative hazard to survival probability
    exp(-Lambda)
  }

  # RMST Contrast

  ## Define the RMST contrast as a function of the model coefficient vector.
  ## This function is subsequently used to obtain the numerical gradient
  ## required for the delta-method standard error.
  rmst.diff <- function(beta) {
```

```

    rmst.fn(t.grid, cb.surv(beta, 1, t.grid)) -
    rmst.fn(t.grid, cb.surv(beta, 0, t.grid))
  }

  ## Evaluate the RMST contrast at the estimated model coefficients
  cb.rmst <- rmst.diff(beta.hat)

  # Numerical Gradient

  ## Calculate the numerical gradient of the RMST contrast with respect to
  ## each model coefficient using a central finite-difference approximation
  p <- length(beta.hat)
  m <- length(t.grid)

  grad <- matrix(NA_real_, nrow = m, ncol = p)
  colnames(grad) <- names(beta.hat)

  ## Step size used for the finite-difference approximation
  eps <- 1e-6

  ## Perturb each coefficient in turn and approximate the corresponding
  ## derivative of the RMST contrast
  for (j in seq_len(p)) {
    bp <- bm <- beta.hat
    bp[j] <- bp[j] + eps
    bm[j] <- bm[j] - eps

    grad[, j] <- (rmst.diff(bp) - rmst.diff(bm)) / (2 * eps)
  }

  # Delta-Method Standard Error

  ## Align the model covariance matrix with the coefficient ordering used by
  ## the numerical gradient before applying the delta-method variance formula
  cb.var <- cb.var[colnames(grad), colnames(grad), drop = FALSE]

  ## Calculate the RMST standard error at each restriction time using
  ##  $\text{Var}\{g(\beta)\} = \text{grad}' \text{Var}(\beta) \text{grad}$ 
  cb.rmst.se <- vapply(
    seq_len(m),
    function(i) {
      g <- grad[i, ]
      sqrt(drop(t(g) %*% cb.var %*% g))
    },
    numeric(1)
  )

  return(list(
    cb.rmst = cb.rmst,
    cb.rmst.se = cb.rmst.se
  ))
}

```

Counterfactual Survival Curves: MS-Cox and IPW Case-Base Models

MS-Cox Counterfactual Survival Curves

```
## Estimate counterfactual survival under both treatment levels using the
## marginal structural Cox model. A = 0 corresponds to no RHC, while A = 1
## corresponds to RHC.
sf.cox <- survfit(
  model.5,
  newdata = survival.curves.data
)

## Interpolate the estimated survival curve under no RHC onto the common
## time grid used for subsequent comparison and RMST calculation
S0.cox <- approx(
  sf.cox$time,
  sf.cox$surv[, 1],
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

## Interpolate the corresponding lower 95% confidence limit under no RHC
S0.lower.cox <- approx(
  sf.cox$time,
  sf.cox$lower[, 1],
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

## Interpolate the corresponding upper 95% confidence limit under no RHC
S0.upper.cox <- approx(
  sf.cox$time,
  sf.cox$upper[, 1],
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

## Interpolate the estimated survival curve under RHC onto the common time grid
S1.cox <- approx(
  sf.cox$time,
  sf.cox$surv[, 2],
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

## Interpolate the corresponding lower 95% confidence limit under RHC
```

```

S1.lower.cox <- approx(
  sf.cox$time,
  sf.cox$lower[, 2],
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

## Interpolate the corresponding upper 95% confidence limit under RHC
S1.upper.cox <- approx(
  sf.cox$time,
  sf.cox$upper[, 2],
  xout = tau.grid,
  method = "constant",
  rule = 2
)$y

```

IPW Case-Base Counterfactual Survival Curves

```
## Estimate counterfactual survival under no RHC and RHC using the IPW
## case-base model over the common time grid
Sf.cb <- absoluteRisk(
  object = model.8,
  time = tau.grid,
  newdata = survival.curves.data,
  type = "survival"
)

## Extract the estimated survival curves for no RHC and RHC, respectively
S0.cb <- Sf.cb[, 2]
S1.cb <- Sf.cb[, 3]

# Bootstrap Confidence Intervals for the IPW Case-Base Curves

## Use nonparametric bootstrap resampling to obtain pointwise confidence
## intervals for the counterfactual survival curves estimated by the IPW
## case-base model
set.seed(s)
n.boot <- 200

## For each bootstrap sample, refit the IPW case-base model and obtain
## counterfactual survival estimates under both treatment levels
boot.surv <- replicate(n.boot, {
  idx <- sample(nrow(data.set), replace = TRUE)
  boot.data <- data.set[idx, ]

  boot.fit <- FSH(
    Delta ~ pspline(time, df = 2) * A,
    data = boot.data,
    ratio = 10,
    w = ipw.stabilized[idx]
  )

  ar <- absoluteRisk(
    boot.fit,
    time = tau.grid,
    newdata = survival.curves.data,
    type = "survival"
  )

  list(
    S0 = ar[, 2],
    S1 = ar[, 3]
  )
})

## Organize the bootstrap estimates into separate matrices for the no-RHC
## and RHC counterfactual survival curves
S0.boot <- sapply(1:n.boot, function(i) boot.surv[["S0", i]])
S1.boot <- sapply(1:n.boot, function(i) boot.surv[["S1", i]])
```

```

## Calculate the pointwise lower 95% bootstrap confidence limit for the
## no-RHC counterfactual survival curve
S0.lower.cb <- apply(
  S0.boot,
  1,
  quantile,
  probs = 0.025,
  na.rm = TRUE
)

## Calculate the pointwise upper 95% bootstrap confidence limit for the
## no-RHC counterfactual survival curve
S0.upper.cb <- apply(
  S0.boot,
  1,
  quantile,
  probs = 0.975,
  na.rm = TRUE
)

## Calculate the pointwise lower 95% bootstrap confidence limit for the
## RHC counterfactual survival curve
S1.lower.cb <- apply(
  S1.boot,
  1,
  quantile,
  probs = 0.025,
  na.rm = TRUE
)

## Calculate the pointwise upper 95% bootstrap confidence limit for the
## RHC counterfactual survival curve
S1.upper.cb <- apply(
  S1.boot,
  1,
  quantile,
  probs = 0.975,
  na.rm = TRUE
)

```

Head-to-Head Comparison of Counterfactual Survival Curves

```

## Compare the MS-Cox and IPW case-base estimates separately under each
## treatment condition. The confidence bands are plotted first so that the
## survival curves remain visible on top of the shaded regions.

# Counterfactual Survival Under No RHC

## Initialize the plotting region for the counterfactual survival curves
## under no RHC (A = 0)

```

```

plot(
  tau.grid, S0.cox,
  type = "n",
  lwd = 2,
  col = col.cox,
  lty = 1,
  ylim = c(0, 1),
  xlab = "Time",
  ylab = "Survival Probability",
  main = "Counterfactual Survival Curves (No RHC)"
)

## Add the 95% confidence band for the MS-Cox survival estimate
polygon(
  c(tau.grid, rev(tau.grid)),
  c(S0.lower.cox, rev(S0.upper.cox)),
  col = fill.cox,
  border = NA
)

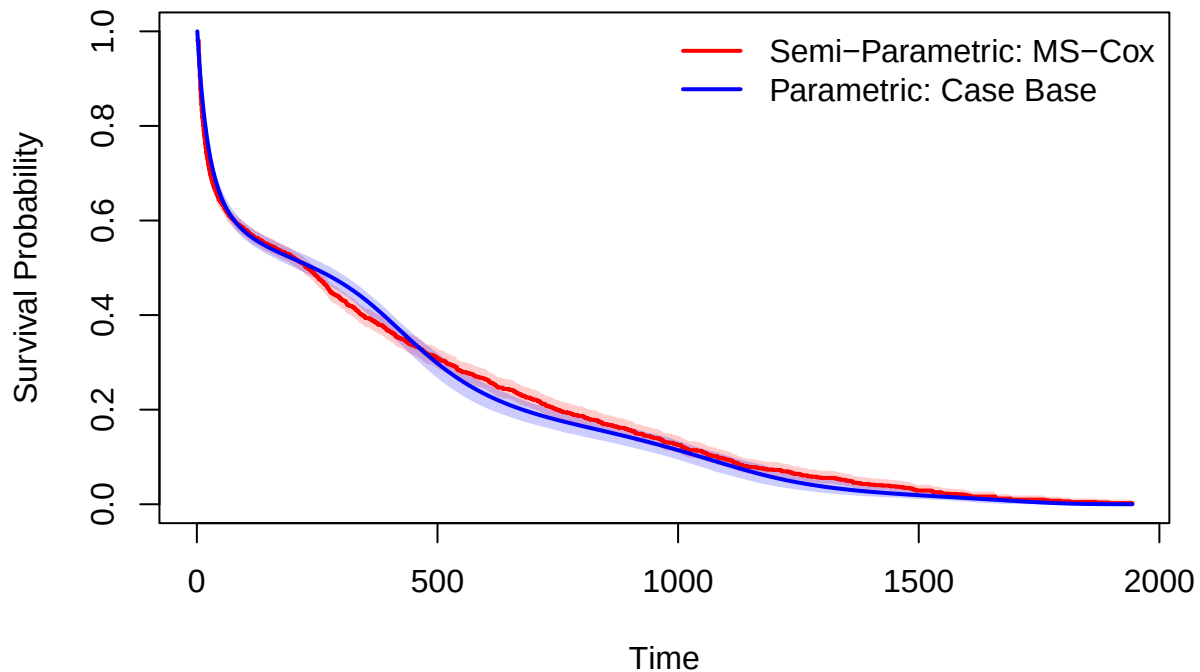
## Add the 95% bootstrap confidence band for the IPW case-base survival
## estimate
polygon(
  c(tau.grid, rev(tau.grid)),
  c(S0.lower.cb, rev(S0.upper.cb)),
  col = fill.cb,
  border = NA
)

## Overlay the estimated survival curves. The MS-Cox curve is a step function,
## whereas the case-base model produces a smooth survival curve.
lines(S0.cox, type = "s", lwd = 2, col = col.cox, lty = 1)
lines(S0.cb, type = "l", lwd = 2, col = col.cb, lty = 1)

## Identify the two estimation approaches in the comparison plot
legend(
  "topright",
  legend = c(
    "Semi-Parametric: MS-Cox",
    "Parametric: Case Base"
  ),
  col = c(
    col.cox, col.cb
  ),
  lty = c(1, 1),
  lwd = 2,
  bty = "n"
)

```

Counterfactual Survival Curves (No RHC)



```
# Counterfactual Survival Under RHC

## Initialize the plotting region for the counterfactual survival curves
## under RHC (A = 1)
plot(
  tau.grid, S1.cox,
  type = "n",
  lwd = 2,
  col = col.cox,
  lty = 1,
  ylim = c(0, 1),
  xlab = "Time",
  ylab = "Survival Probability",
  main = "Counterfactual Survival Curves (RHC)"
)

## Add the 95% confidence band for the MS-Cox survival estimate
polygon(
  c(tau.grid, rev(tau.grid)),
  c(S1.lower.cox, rev(S1.upper.cox)),
  col = fill.cox,
  border = NA
)

## Add the 95% bootstrap confidence band for the IPW case-base survival
## estimate
```

```

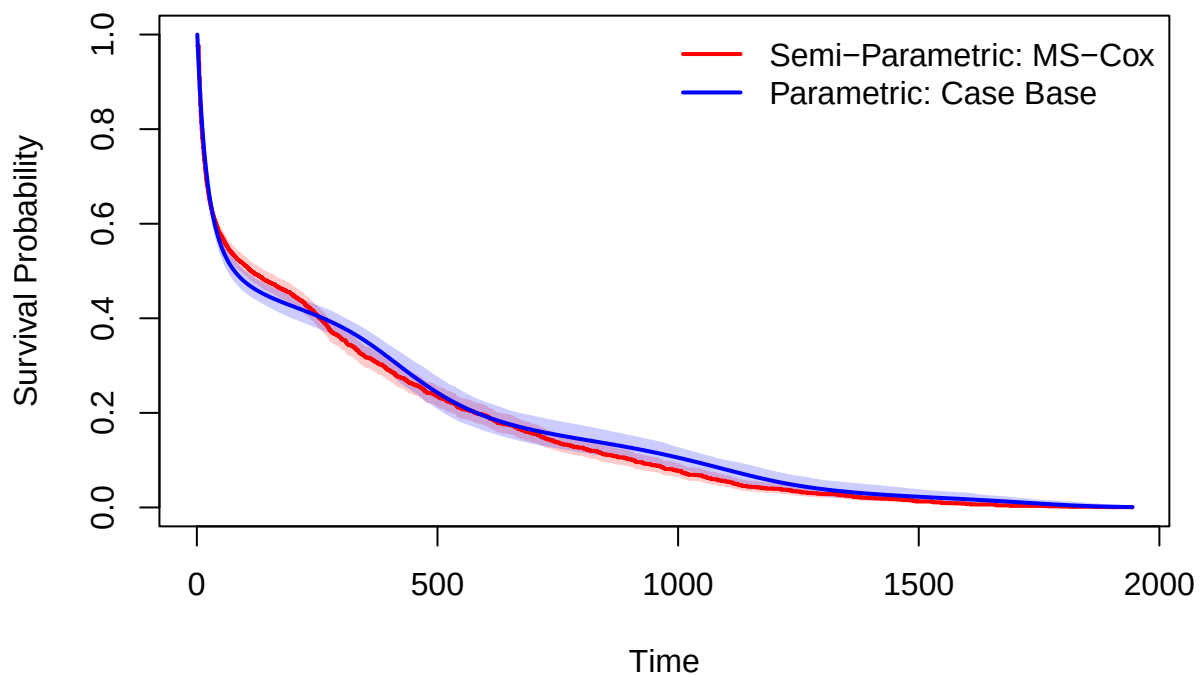
polygon(
  c(tau.grid, rev(tau.grid)),
  c(S1.lower.cb, rev(S1.upper.cb)),
  col = fill.cb,
  border = NA
)

## Overlay the estimated survival curves under RHC
lines(S1.cox, type = "s", lwd = 2, col = col.cox, lty = 1)
lines(S1.cb, type = "l", lwd = 2, col = col.cb, lty = 1)

## Identify the two estimation approaches in the comparison plot
legend(
  "topright",
  legend = c(
    "Semi-Parametric: MS-Cox",
    "Parametric: Case Base"
  ),
  col = c(
    col.cox, col.cb
  ),
  lty = c(1, 1),
  lwd = 2,
  bty = "n"
)

```

Counterfactual Survival Curves (RHC)



RMST Comparison: MS-Cox vs. IPW Case-Base

RMST Estimates and Standard Errors

```
## Extract the robust standard error of the treatment log-hazard coefficient
## from the marginal structural Cox model
model.5.psi.se <- sqrt(diag(vcov(model.5))["A"])

## Obtain the heteroskedasticity-consistent covariance matrix for the IPW
## case-base model. This covariance matrix is used in the numerical
## delta-method calculation of the RMST standard error.
model.8.var <- cb.var <- vcovHC(model.8, type = "HCO")

## Calculate the RMST contrast and its delta-method standard error for the
## marginal structural Cox model
cox.rmst.list <- cox.rmst.list(
  model.5, tau.grid, model.5.psi.se, survival.curves.data
)

## Calculate the RMST contrast and its delta-method standard error for the
## IPW case-base model
cb.rmst.list <- cb.rmst.list(
  model.8, model.8.var, tau.grid, constant.HR = TRUE
)

# Extract RMST Estimates and Standard Errors

## Extract the estimated RMST contrasts and their corresponding standard errors
## from both estimation approaches
cox.rmst <- cox.rmst.list$cox.rmst
cox.rmst.se <- cox.rmst.list$cox.rmst.se

cb.rmst <- cb.rmst.list$cb.rmst
cb.rmst.se <- cb.rmst.list$cb.rmst.se

# Pointwise 95% Confidence Intervals

## Construct pointwise 95% Wald confidence intervals for the MS-Cox RMST
## contrast over the range of restriction times
cox.lower <- cox.rmst - 1.96 * cox.rmst.se
cox.upper <- cox.rmst + 1.96 * cox.rmst.se

## Construct pointwise 95% Wald confidence intervals for the IPW case-base
## RMST contrast over the range of restriction times
cb.lower <- cb.rmst - 1.96 * cb.rmst.se
cb.upper <- cb.rmst + 1.96 * cb.rmst.se
```

RMST Comparison Plot

```
## Determine a common y-axis range that accommodates the point estimates and
## confidence intervals from both estimation approaches
ylim <- range(
  c(
    cox.lower, cox.upper,
    cb.lower, cb.upper
  ),
  na.rm = TRUE
)

## Initialize the plotting region for comparing the estimated causal RMST
## contrasts over the full range of restriction times
plot(
  tau.grid, cox.rmst,
  type = "n",
  ylim = ylim,
  xlab = expression(tau),
  ylab = "RMST Difference",
  main = "Causal RMST up to Tau"
)

# 95% Confidence Bands

## Add the pointwise 95% confidence band for the MS-Cox RMST contrast
polygon(
  c(tau.grid, rev(tau.grid)),
  c(cox.lower, rev(cox.upper)),
  col = adjustcolor(col.cox, alpha.f = 0.2),
  border = NA
)

## Add the pointwise 95% confidence band for the IPW case-base RMST contrast
polygon(
  c(tau.grid, rev(tau.grid)),
  c(cb.lower, rev(cb.upper)),
  col = adjustcolor(col.cb, alpha.f = 0.2),
  border = NA
)

# RMST Contrast Estimates

## Overlay the estimated causal RMST contrast from the MS-Cox model
lines(tau.grid, cox.rmst, col = col.cox, lwd = 2)

## Overlay the estimated causal RMST contrast from the IPW case-base model
lines(tau.grid, cb.rmst, col = col.cb, lwd = 2)

## Add a reference line at an RMST difference of zero, corresponding to equal
## restricted mean survival under the two treatment conditions
abline(h = 0, lty = 3)
```

```

# Model Legend

## Identify the two estimation approaches in the comparison plot
legend(
  "bottomleft",
  legend = c("MS-Cox", "IPW Case-Base"),
  col = c(col.cox, col.cb),
  lwd = 2,
  bty = "n"
)

```

